IGCSE 0607 Practice Questions — E8.3

Magnitude of a vector

Total Questions: 6 — **Total Marks:** 38 Marks

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1 Given the vectors $\mathbf{p} = \begin{pmatrix} -8 \\ 15 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$.

(a) Calculate $|\mathbf{p}|$.

..... **[2]**

(b) Calculate $|\mathbf{p} + 2\mathbf{q}|$.

..... **[3]**

2 Points A, B, and C have coordinates $(3, -2)$, $(-5, y)$, and $(7, 1)$ respectively.

(a) Calculate $|\vec{AC}|$.

..... **[2]**

(b) The magnitude of vector $\vec{AB}$ is 17.
Work out the two possible values of y.

$y =$ ..... , $y =$ ..... **[5]**

3 Given the vectors $\mathbf{u} = \begin{pmatrix} 2k \\ k-3 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} k+1 \\ 2 \end{pmatrix}$.

(a) Given that $|\mathbf{u} + \mathbf{v}| = \sqrt{50}$, show that $5k^2 + 2k - 24 = 0$.

[4]

(b) Solve the equation

$$5k^2 + 2k - 24 = 0$$

$k =$ ..... , $k =$ ..... **[4]**

(c) Hence, find the two possible column vectors for $\mathbf{u}$.

$\mathbf{u} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ ..... , $\mathbf{u} = \begin{pmatrix} \quad \\ \quad \end{pmatrix}$ ..... **[2]**

4 Vector $\mathbf{v}$ is given as

$$\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$$

Work out $|\mathbf{v}|$, the magnitude of vector $\mathbf{v}$.

..... **[2]**

5 **(a)** The points A and B have coordinates $(2, -3)$ and $(9, 4)$ respectively.
Work out the column vector $\vec{AB}$.

$\begin{pmatrix} \quad \\ \quad \end{pmatrix}$ **[2]**

(b) Calculate $|\vec{AB}|$.
Give your answer correct to 3 significant figures.

..... **[2]**

(c) Vector $\mathbf{p} = \begin{pmatrix} k \\ 4 \end{pmatrix}$.
Given that $|\mathbf{p}| = 5$, find the two possible values of k.

$k =$ ..... , $k =$ ..... **[3]**

6 Vector $\mathbf{u} = \begin{pmatrix} x \\ x+2 \end{pmatrix}$.

(a) The magnitude of vector **u** is 10.
Show that $x^2 + 2x - 48 = 0$.

[3]

(b) Solve the equation

$$x^2 + 2x - 48 = 0$$

$x =$ ..... , $x =$ ..... **[2]**

(c) Hence write down the two possible column vectors for **u**.

[2]